

From Research to Reality: Building GUI Agents That Actually Work

Resources:

<https://www.youtube.com/@voxel51>

Welcome to the Visual Agents Workshop Series, your virtual pass to learn about visual agents - how they work, how to develop them and how to fine-tune them.

From <https://voxel51.com/events/from-research-to-reality-building-gui-agents-that-actually-work-august-15-2025>

Part 1: Navigating the GUI Agent Landscape

Understanding the Foundation Before Building

The GUI agent field is evolving rapidly, but success requires an understanding of what came before. In this opening session, we'll map the terrain of GUI agent research—from the early days of MiniWoB's simplified environments to today's complex, multimodal systems tackling real-world applications. You'll discover why standard vision models fail catastrophically on GUI tasks, explore the annotation bottlenecks that make GUI datasets so expensive to create, and understand the platform fragmentation that makes "click a button" mean twenty different things across datasets.

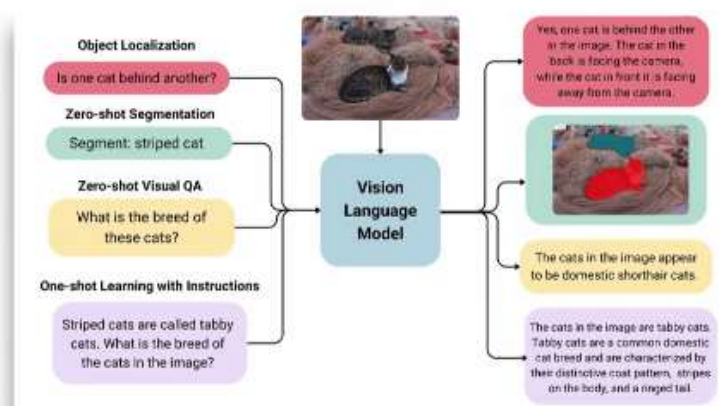
We'll dissect the most influential datasets (Mind2Web, AITW, Rico) and models that have shaped the field, examining their strengths, limitations, and the research gaps they reveal. By the end, you'll have a clear picture of where GUI agents excel, where they struggle, and, most importantly, where the opportunities lie for your own contributions.

myMeetingNotes

https://github.com/harpreetsahota204/gui_agent_research_landscape

Visual Agents vs Vision Language Models

- **General Vision-Language Models:** Understanding and describing visual content
- **Visual Agents (Based on VLMs):** Taking actions based on visual understanding



Source: Hugging Face Blog

Required capabilities for a GUI agent

- **UI Grounding:** Locating and identifying interactive elements
- **OCR (Optical Character Recognition):** Extracting text from visual elements
- **Screen Understanding:** Interpreting the content and layout of the entire screen
- **Action Prediction:** Determining the next appropriate action

Teaching AI where to click is really hard

- AI must convert visual understanding into exact pixel coordinates without direct spatial training
- Current systems demand one perfect click point but buttons accept clicks anywhere
- Models only get **binary** right/wrong feedback instead of **"close but missed"** guidance
- GUI-Actor's attention approach focuses on screen regions rather than guessing coordinates
- New methods accept multiple valid click locations and provide gradual learning feedback

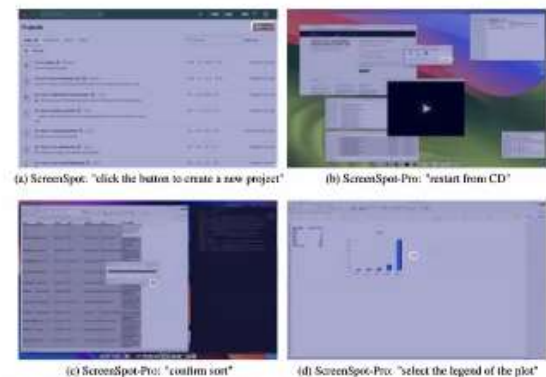


Figure 5: Example visualizations from (a) ScreenSpot and (b)(c)(d) ScreenSpot-Pro. Each image shows the original interface with an overlaid attention map indicating regions of interest of GUI-Actor. The attention maps largely overlap with the ground truth areas (red bounding boxes), demonstrating that the model can effectively capture the accurate UI elements.

GUI_Actor: Coordinate-Free Visual Grounding for GUI Agents; arXiv 2506.03143

Get the slides here 📌



Next: Part 2

<https://voxel51.com/events/from-research-to-reality-building-gui-agents-that-actually-work-august-22-2025>

Part 2: From Pixels to Predictions - Building Your GUI Dataset

Hands-On Dataset Creation and Curation with FiftyOne

The best GUI models are only as good as their training data, and the best datasets are built by understanding what makes GUI interactions fundamentally different from natural images. In this practical session, you'll build a complete GUI dataset from scratch, learning to capture the precise annotations that GUI agents need.

Using FiftyOne as your data management backbone, you'll import diverse GUI screenshots, explore annotation strategies that go beyond bounding boxes, and implement efficient labeling workflows. We'll tackle the real challenges: handling platform differences, managing annotation quality, and creating datasets that transfer to new domains. You'll also learn advanced techniques like synthetic data generation and automated prelabeling to scale your annotation efforts.

Walk away with a production-ready dataset and the skills to build more—because in GUI agents, data quality determines everything.

By the end, you'll have both a dataset and the methodology to build the next generation of GUI training data.

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